6. *Message decoding*. The message decoder translates the encoded message to a JSON object to be published for ingestion and storage.

This flow of data from my garden tap, to its eventual destination where it is decoded is shown in figure 3.12.

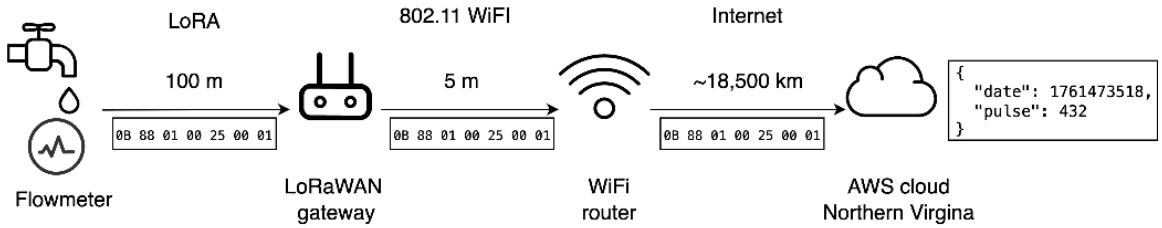


Figure 3.12 The route travelled by a reading from the LoRaWAN flowmeter attached to my garden tap.

This same flow applies whether the data originates from an electronic LoRaWAN sensor or manual pool chemistry testing—the architecture abstracts away the collection and communication details to provide a consistent interface for digital twin logic.

3.7.6 Architectural principles

Several key principles emerge from this architecture that apply to digital twin implementations at any scale.

1. *Embrace diversity*. Rather than forcing all data collection into a single technology, design systems that can accommodate multiple approaches based on practical constraints.
2. *Converge on standards*. While collection methods may vary, standardize on common protocols and data formats for transport and processing to simplify application development.
3. *Plan for evolution*. Design loosely-coupled interfaces that can accommodate new sensor types and communication methods without requiring changes to core application logic.
4. *Balance automation and pragmatism*. Automate what provides clear value, but don’t hesitate to use manual collection when it’s more cost-effective or reliable.